Basics 6: Linear Algebra and Eigenvalue Problems

I - Mathematical and computational foundations

Vector spaces, inner products, generalized eigenproblems, conditioning, and physical modes

Textbook draft, version 1.0. Discretized plasma equations become algebraic systems. Equilibria require solves, waves require eigenpairs, and multiphysics coupling requires stable linearization and conditioning. This chapter is designed for a reader with no prior plasma-physics training. It distinguishes exact identities from physical approximations and connects the central equations to later simulation modules.

Contents

1 Learning objectives

After completing this note, the reader should be able to:

- Interpret arrays and functions as vectors in a vector space.
- Use bases, inner products, norms, and orthogonality.
- Explain rank, null spaces, solvability, and condition number.
- Derive ordinary and generalized eigenvalue problems.
- Use symmetry, positivity, and Rayleigh quotients.
- Assess residuals, normalization, convergence, and mode tracking.

2 Prerequisites and reading strategy

- Basics 1 and elementary algebra; integration for continuous inner products.

On a first reading, emphasize physical meaning and dimensional checks. On a second reading, reproduce the derivations. Attempt the computational laboratory only after the limiting cases are understood.

3 Vector spaces and bases

A vector space is closed under addition and scalar multiplication. Functions form vector spaces as naturally as finite arrays. In a basis $\{\mathbf{e}_i\}$,

$$\mathbf{x} = \sum_i x_i \mathbf{e}_i. \tag{3.1}$$

Discretization represents a continuous field by a finite coordinate vector. The basis may be grid values, finite-element functions, Fourier modes, or learned features.

4 Inner products and orthogonality

The Euclidean inner product is $\mathbf{x}^T\mathbf{y}$. A weighted inner product is

$$\langle \mathbf{x}, \mathbf{y} \rangle_M = \mathbf{x}^T M \mathbf{y}, \tag{4.1}$$

with symmetric positive-definite M. A continuous analogue is

$$\langle \xi, \eta \rangle = \int w(s)\xi(s)\eta(s)\, ds. \tag{4.2}$$

Orthogonality depends on the chosen metric. Generalized eigenmodes are often orthogonal in a mass-matrix metric rather than the ordinary dot product.

5 Linear systems, null spaces, and conditioning

A square system $A\mathbf{x} = \mathbf{b}$ has a unique solution if A is nonsingular. The null space contains solutions of $A\mathbf{x} = 0$. Gauge freedoms, pure-Neumann boundary conditions, and unconstrained normalizations often create null spaces.

The condition number

$$\kappa(A) = \|A\| \left\| A^{-1} \right\| \tag{5.1}$$

measures worst-case amplification of input error. Nondimensionalization, scaling, and preconditioning improve conditioning without changing the intended physical solution.

6 Eigenvalues and dynamical modes

An eigenpair satisfies

$$A\mathbf{x} = \lambda\mathbf{x}, \qquad \mathbf{x} \neq 0. \tag{6.1}$$

For $\dot{\mathbf{q}} = A\mathbf{q}$, a modal solution $\mathbf{q} = \mathbf{x}e^{\lambda t}$ has growth and oscillation set by λ.

A second-order system

$$M\ddot{\mathbf{q}} + K\mathbf{q} = 0 \tag{6.2}$$

with $\mathbf{q} = \mathbf{x}e^{-i\omega t}$ gives

$$K\mathbf{x} = \omega^2 M\mathbf{x}. \tag{6.3}$$

K represents restoring forces and M inertia. Module 4 uses this structure.

7 Symmetry, positivity, and Rayleigh quotients

If K and M are symmetric and M is positive definite, generalized eigenvalues are real under appropriate boundary conditions. The Rayleigh quotient

$$\mathcal{R}(\mathbf{x}) = \frac{\mathbf{x}^T K\mathbf{x}}{\mathbf{x}^T M\mathbf{x}} \tag{7.1}$$

equals ω^2 at an eigenvector. Positive K implies positive squared frequencies. An indefinite force operator may produce negative ω^2, signaling ideal instability.

8 Residuals, normalization, sparse solvers, and mode identity

A computed eigenpair should satisfy a normalized residual such as

$$r = \frac{\|K\mathbf{x} - \lambda M\mathbf{x}\|}{\|K\mathbf{x}\| + |\lambda|\|M\mathbf{x}\|}. \tag{8.1}$$

A small residual proves algebraic consistency, not continuum convergence. Eigenvectors have arbitrary scale and sign, so impose

$$\mathbf{x}^T M\mathbf{x} = 1 \tag{8.2}$$

and a reproducible sign rule.

Sparse Krylov and shift-invert methods find selected modes without diagonalizing the whole matrix. During parameter scans, frequency sorting can swap branch identity near avoided crossings. Use metric overlaps, harmonic weights, and localization.

9 Computational laboratory

Assemble mass and stiffness matrices for a fixed-end mass-spring chain. Solve $Kx = \omega^2 Mx$, normalize in the M inner product, and verify orthogonality and residuals. Refine the chain and compare with the analytic string limit. Perturb one spring and track modes by overlap rather than eigenvalue rank.

10 Summary

- Discretization maps fields into finite-dimensional vector spaces.
- Inner products define normalization and orthogonality.
- Generalized eigenproblems separate inertia and restoring forces.

- Symmetry and positive definiteness strongly constrain the spectrum.
- Residuals, refinement, and mode tracking are all required.

Symbols and acronyms used in this document

Symbol / acronym	Name	Meaning in this document
A, K, M	operators	Generic, stiffness, and mass matrices.
λ, ω^2	eigenvalue	Modal growth parameter or squared frequency.
$\mathbf{x}$	eigenvector	Discrete mode shape.
$\kappa(A)$	condition number	Sensitivity of a linear solve.
$\mathcal{R}$	Rayleigh quotient	Energy ratio estimating an eigenvalue.

11 Exercises

1. Prove orthogonality of distinct eigenvectors of a real symmetric matrix.
2. Derive the generalized Rayleigh quotient.
3. Give an example where a small matrix residual does not imply a good physical solution.
4. Explain why eigenvector scale and sign are arbitrary.
5. Derive the eigenvalues of a coupled 2×2 matrix.

12 References

References

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